

Department of Physics, FNSPE, CTU in Prague  
02YELMA - Electricity and Magnetism  
**Midterm 2 (B)**

15.5.2026 / 16:00 - 17:30

**Q1:** The magnetic field of a monochromatic electromagnetic wave propagating in free space is given by  $\vec{B}(x, t) = B_0 \cos(kx - \omega t) \hat{z}$

- (a) Find the instantaneous power transmitted through a circular area of radius  $b$  in the plane perpendicular to the direction of propagation.
- (b) If the wave passes through an ideal polarizer oriented at an angle  $\pi/4$  with respect to the direction of electric field polarization, find the amplitude of the transmitted electric field.

**Q2:** A long, straight, solid cylindrical conductor of radius  $a$  carries a steady total current  $I_0$  in the  $+\hat{x}$  direction. The volume current density varies with radial distance  $r$  from the axis as  $\vec{J}(r) = J_1 \left(\frac{r}{a}\right) \hat{x}$  for  $0 \leq r \leq a$  and  $\vec{J}(r) = 0$  for  $r > a$ . Express  $J_1$  in terms of  $I_0$  and  $a$ , and then find the magnetic field  $\vec{B}(r)$  for  $r < a$  and  $r > a$ .

**Q3:** A metallic cylinder of radius  $b$  and height  $H$  carries a total charge  $Q$ . It rotates about its symmetry axis  $\hat{z}$  with a constant angular velocity  $\omega$ . Calculate the magnetic dipole moment of the cylinder. — *Assume the charge is uniformly distributed over the curved outer surface.*

**Q4:** A parallel-plate capacitor with circular plates of radius  $a$  is being charged by a steady current  $I_0$ . The plate separation is  $d$ , with  $d \ll a$ , so edge effects may be neglected. Assume vacuum between the plates. At a given instant, the electric field between the plates is uniform and directed along the  $x$ -axis  $\vec{E}(t) = E(t) \hat{x}$ . Using Poynting's theorem, determine the energy stored in the capacitor at a given time. — *Solutions without using Poynting's theorem will not receive credit.*

Maxwell's equations

$$\oint_S \vec{E} \cdot d\vec{a} = \frac{Q_{\text{enc}}}{\epsilon_0}$$

$$\oint_S \vec{B} \cdot d\vec{a} = 0$$

$$\oint_C \vec{E} \cdot d\vec{\ell} = -\frac{d}{dt} \int_S \vec{B} \cdot d\vec{a}$$

$$\oint_C \vec{B} \cdot d\vec{\ell} = \mu_0 I_{\text{enc}} + \mu_0 \epsilon_0 \frac{d}{dt} \int_S \vec{E} \cdot d\vec{a}$$

$$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0}$$

$$\nabla \cdot \vec{B} = 0$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\nabla \times \vec{B} = \mu_0 \vec{J} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t}$$